Arfat Naik

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Professional Summary

Results-driven Data Science and Machine Learning student with hands-on experience building, training, and deploying end-to-end ML systems on real-world datasets. Proficient in Python, Scikit-learn, TensorFlow, Pandas, and Flask. Experienced in supervised learning, feature engineering, NLP-based recommendation systems, and automated ML pipelines. Seeking a Data Science or Machine Learning internship to apply analytical and modeling skills to impactful problems.

Technical Skills

Languages: Python (NumPy, Pandas, Scikit-learn, TensorFlow, Joblib), SQL, JavaScript
Machine Learning: Supervised Learning (Regression, Classification), Feature Engineering, Model Training & Evaluation, Cross-Validation, Hyperparameter Tuning, Class Imbalance Handling
Performance Metrics: ROC-AUC, Precision, Recall, F1-Score, R^2, RMSE
NLP & Recommendation: TF-IDF Vectorization, Cosine Similarity, Content-Based Filtering, Sparse Matrix Optimization
Data Science: EDA, Data Visualization (Matplotlib, Seaborn, Chart.js), Statistical Analysis, Data Cleaning, Outlier Detection, Missing Value Imputation
MLOps & Tools: Flask (Model Serving), GitHub Actions (CI/CD), Redis, PostgreSQL (Supabase), Git, Jupyter Notebook, VS Code

Projects

AtmosIQ - Air Quality Forecasting System | *Python, Flask, Redis, Supabase, GitHub Actions* Live | GitHub

- Developed an end-to-end **time-series forecasting system** using a **Random Forest regression model** to predict AQI 3 hours ahead, attaining $\mathbf{R^2 \approx 0.85}$ through engineered lag features, rolling window statistics, and meteorological variables.
- Built a fully automated **hourly ML inference pipeline** with GitHub Actions, Redis (real-time caching), and Supabase (PostgreSQL), enabling continuous data ingestion, feature transformation, prediction, and storage with zero manual steps.
- Deployed a **production Flask dashboard** on Railway with Chart.js-powered visualizations for PM2.5, PM10, current AQI, and 3-hour forecasts, translating ML outputs into actionable environmental insights.

Credit Risk Prediction System | *Python, Logistic Regression, Scikit-learn, Flask* Live | GitHub

- Designed and trained a **Logistic Regression credit risk model** with **cost-sensitive thresholding** to maximize recall on loan default detection while minimizing false approvals.
- Applied rigorous **data preprocessing**: missing value imputation, outlier handling, feature engineering, train-test leakage prevention, and **class imbalance management** for robust and unbiased model evaluation.
- Deployed model as a **Flask REST API** providing real-time default probability scores and binary decisions, making predictions accessible through a live web interface on Vercel.

Movie Recommendation System | *Python, TF-IDF, Cosine Similarity, Flask, NLP* Live | GitHub

- Built a **content-based NLP recommendation engine** for 5,000+ TMDB movies, using **TF-IDF vectorization and cosine similarity** on a multi-field feature corpus (genres, keywords, cast, director, overview).
- Optimized similarity computation via **sparse matrix representations**, reducing runtime by **60%** and enabling real-time recommendations via a deployed Flask API with input validation.

Education

Rizvi College of Engineering Mumbai, India
Bachelor of Engineering in Electronics and Computer Engineering *2023 – 2027*

- CGPA: 7.27/10 | Semester 6
- Relevant Coursework: Machine Learning, Statistics, Data Structures & Algorithms, Database Management Systems, Linear Algebra

Certifications

Programming, Data Structures and Algorithms using Python | *NPTEL* Certificate